

Program

Server.java

```
import java.io.*;
import java.net.*;

public class server {

    public static void main(String[] args) throws Exception {

        ServerSocket server = new ServerSocket(9002);
        System.out.println("Server waiting for connection...");

        Socket socket = server.accept();
        System.out.println("Client Connected.");

        ObjectInputStream in = new ObjectInputStream(socket.getInputStream());

        int n = in.readInt();

        int[][] matrix = (int[][]) in.readObject();

        System.out.println("\nReceived Matrix:");

        for(int i=0;i<n;i++)
```

```
{  
    for(int j=0;j<n;j++)  
        System.out.print(matrix[i][j]+" ");  
    System.out.println();  
}
```

```
boolean upper=true;  
boolean lower=true;  
boolean diagonal=true;
```

```
for(int i=0;i<n;i++)  
{  
    for(int j=0;j<n;j++)  
    {  
        if(i>j && matrix[i][j]!=0)  
            upper=false;  
  
        if(i<j && matrix[i][j]!=0)  
            lower=false;  
  
        if(i!=j && matrix[i][j]!=0)  
            diagonal=false;  
    }  
}
```

```
String result;
```

```
        if(diagonal)
            result="Diagonal Matrix";
        else if(upper)
            result="Upper Triangular Matrix";
        else if(lower)
            result="Lower Triangular Matrix";
        else
            result="Normal Matrix";

        System.out.println("\nMatrix Type : "+result);

        DataOutputStream out =
            new DataOutputStream(socket.getOutputStream());

        out.writeUTF(result);

        socket.close();
        server.close();
    }
}
```

Client.java

```
import java.io.*;
import java.net.*;
import java.util.*;
```

```
public class client {

    public static void main(String[] args) throws Exception {

        Socket socket = new Socket("localhost",9002);

        Scanner sc = new Scanner(System.in);

        Random rand = new Random();

        System.out.print("Enter order of matrix : ");
        int n=sc.nextInt();

        int[][] matrix=new int[n][n];

        for(int i=0;i<n;i++)
        {
            for(int j=0;j<n;j++)
            {
                matrix[i][j]=rand.nextInt(50)+1;
            }
        }

        int type=rand.nextInt(4);

        switch(type)
```

```
{  
    case 0:    //Upper  
        for(int i=0;i<n;i++)  
            for(int j=0;j<n;j++)  
                if(i>j)  
                    matrix[i][j]=0;  
        break;  
  
    case 1:    //Lower  
        for(int i=0;i<n;i++)  
            for(int j=0;j<n;j++)  
                if(i<j)  
                    matrix[i][j]=0;  
        break;  
  
    case 2:    //Diagonal  
        for(int i=0;i<n;i++)  
            for(int j=0;j<n;j++)  
                if(i!=j)  
                    matrix[i][j]=0;  
        break;  
  
    default:   //Normal Matrix  
        break;  
}
```

```
System.out.println("\nGenerated Matrix");
```

```
for(int i=0;i<n;i++)
```

```
{
```

```
    for(int j=0;j<n;j++)
```

```
        System.out.print(matrix[i][j]+" ");
```

```
    System.out.println();
```

```
}
```

```
ObjectOutputStream out =
```

```
    new ObjectOutputStream(socket.getOutputStream());
```

```
out.writeInt(n);
```

```
out.writeObject(matrix);
```

```
out.flush();
```

```
DataInputStream in =
```

```
    new DataInputStream(socket.getInputStream());
```

```
String result=in.readUTF();
```

```
System.out.println("\nMatrix Type : "+result);
```

```
socket.close();
```

```
}
```

```
}
```

